

Project Proposal: Multi-Future Pedestrian Trajectory Forecasting with Intent–Trajectory Consistency

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Overview

Autonomous vehicles must anticipate pedestrian behavior to plan safely, especially near crosswalks, intersections, and occlusions. We propose to study joint pedestrian intention estimation (e.g., crossing vs. not crossing) and future trajectory forecasting on the PIE dataset. As a baseline, we will re-implement an existing pipeline (PIEPredict) that predicts intention and regresses a single future trajectory. Building on this baseline, we propose two algorithmic innovations: (i) multi-future (multi-hypothesis) trajectory prediction that outputs K plausible futures to address one-to-many uncertainty; and (ii) an intent–trajectory consistency regularizer that explicitly aligns the predicted crossing probability with the probability mass assigned to crossing-like trajectory modes. We will evaluate improvements using standard intention metrics (F1/AUC), trajectory metrics (MSE/ADE/FDE), and multi-future metrics (minADE/minFDE), alongside qualitative visualizations in challenging scenes.

1 Introduction

1.1 Problem

We aim to solve two coupled prediction tasks for pedestrians in monocular driving videos: (a) intention estimation (predicting whether a pedestrian will cross the street in the near future) and (b) trajectory prediction (forecasting the pedestrian’s future positions over a horizon such as 1–2 seconds). Inputs may include the past bounding-box sequence, pedestrian-centric image crops, and optional context signals (e.g., ego-vehicle speed). Outputs include a crossing probability and a predicted future trajectory (e.g., bounding-box center sequence).

A key difficulty is that the mapping from past observations to future motion is often **one-to-many**: the same history can lead to multiple plausible futures (stop, continue, begin crossing, etc.). Our project focuses on modeling this uncertainty explicitly while keeping the pipeline comparable to a standard baseline.

1.2 Importance and Interest

Accurate pedestrian intention and motion forecasting is safety-critical in autonomous driving. Incorrect forecasts can cause unsafe maneuvers (late braking, wrong yield decisions) or overly conservative driving (unnecessary stops), particularly in crowded urban scenes and near crosswalks.

We are interested in this problem because it provides a practical testbed for algorithmic modeling choices in sequential prediction: deterministic regression vs. probabilistic/multi-future forecasting, and how to enforce semantic consistency between intention (classification) and motion (trajectory distribution). The PIE dataset and existing baselines provide a reproducible starting point, enabling controlled comparisons for our proposed innovations.

1.3 Current State-of-the-Art

Classical approaches for short-horizon motion prediction include simple kinematic models and filters (e.g., constant velocity / Kalman-style predictors), but these often fail under abrupt behavior changes, occlusions, and human interactions. More recent learning-based approaches use sequence models (e.g., encoder–decoder architectures) to jointly predict intention and future motion, leveraging visual and contextual cues.

However, many baseline implementations still predict a **single** future trajectory via MSE regression. In multi-outcome settings, single-trajectory regression tends to average distinct futures, producing implausible “mean” predictions. Modern forecasting literature emphasizes **multi-hypothesis** outputs and metrics like minADE/minFDE to measure whether the predicted set covers the realized future. Our proposal aligns with this trend, while introducing an explicit coupling between intention probability and trajectory-mode probabilities.

1.4 Re-implementation vs. New Approach

We will first re-implement an existing solution as a strong baseline: the PIEPredict pipeline for intention estimation and deterministic trajectory regression. This ensures we have a working end-to-end system (data loading, training, evaluation, and visualization) and a fair reference for comparisons.

On top of the baseline, we propose a new algorithmic approach consisting of two additions: (i) multi-future trajectory prediction that outputs K candidate future trajectories (and optionally their mixture weights), and (ii) intent–trajectory consistency that aligns the predicted crossing probability with the total probability assigned to crossing-like trajectory modes.

1.5 Why Existing Approaches Are Insufficient; Why Ours Should Work Better

Limitation 1: Deterministic regression under one-to-many futures. When multiple futures are plausible, deterministic MSE regression learns a conditional mean that can lie between modes, yielding unrealistic “averaged” trajectories. Our **multi-future** formulation outputs K hypotheses $\{\hat{\mathbf{y}}^{(k)}\}_{k=1}^K$ to cover distinct plausible outcomes. A simple and stable training objective is the Best-of- K (minimum) loss:

$$\mathcal{L}_{\text{BoK}} = \min_{k \in \{1, \dots, K\}} \left\| \hat{\mathbf{y}}^{(k)} - \mathbf{y} \right\|_2^2,$$

which encourages at least one hypothesis to match the ground-truth future.

Limitation 2: Weak coupling between intention and motion. In many pipelines, intention is predicted and then fed as a feature into trajectory prediction, but this does not prevent inconsistent outputs (e.g., high crossing probability paired with a non-crossing trajectory). We propose an explicit consistency regularizer. Let π_k be the predicted mixture weight for hypothesis k (via softmax), and let $\mathcal{C}$ be the set of crossing-like modes (defined by a geometric rule, e.g., entering roadway region or exceeding a lateral displacement threshold). Let p_{cross} be the intention head output. We enforce:

$$\mathcal{L}_{\text{cons}} = \left| p_{\text{cross}} - \sum_{k \in \mathcal{C}} \pi_k \right|.$$

Intuitively, if the model predicts “crossing,” it must allocate probability mass to crossing-like trajectory modes. We expect this to reduce semantically inconsistent predictions and improve interpretability, especially in ambiguous scenes.

1.6 Evaluation Plan, Expected Results, and Timeline

Evaluation Metrics

Intention (classification): Accuracy, Precision/Recall, F1, ROC-AUC (to handle potential imbalance).

Trajectory (deterministic): MSE on predicted positions (and/or ADE/FDE for center points).

Multi-future (set-based): minADE/minFDE over K hypotheses (best-of- K evaluation), and optional Miss Rate at a threshold.

Baselines and Comparisons

We plan to report:

- **Baseline:** PIEPredict-style deterministic trajectory regression + intention estimation.
- **Ours (Innovation 1):** Multi-future trajectory prediction (Best-of- K training; $K = \{5, 10\}$).
- **Ours (Innovation 1+2):** Multi-future + intent–trajectory consistency regularizer.
- **Ablations:** effect of K ; effect of consistency weight λ_{cons} ; different definitions of crossing-like modes $\mathcal{C}$.

Results We Plan to Show

We will provide (i) quantitative tables for intention and trajectory metrics; (ii) multi-future coverage results (minADE/minFDE) demonstrating reduced “averaging” errors; and (iii) qualitative visualizations of K predicted futures with mixture weights, highlighting improvements in ambiguous cases (e.g., near crosswalks, occlusions, groups).

Timeline

- **Week 1-2: Baseline setup and reproduction.** Read the PIE paper, set up the PIE dataset pipeline and the PIEPredict codebase. Run a small end-to-end experiment (train/infer on a subset) to verify that data loading, training, and evaluation scripts work. Reproduce the deterministic baseline (intention classification + single-trajectory regression) and lock down the evaluation protocol: intention metrics (F1/AUC), trajectory metrics (MSE/ADE/FDE), and visualization format
- **Week 3-4: Innovation 1 and Midterm deliverables.** Implement the K -hypothesis trajectory head and train with Best-of- K loss. Validate minADE/minFDE computation and run quick sweeps over $K \in \{5, 10\}$. Write the **midterm report** summarizing baseline results, preliminary quantitative results, and next-step plan.
- **Week 5-6: Innovation 2 (intent–trajectory consistency) + Ablations.** Add mixture weights π_k (softmax) and implement the intent–trajectory consistency regularizer. Define crossing-like mode set $\mathcal{C}$ (e.g., roadway entry or lateral displacement threshold) and tune the consistency weight λ_{cons} . Also run some ablations: (i) baseline vs multi-future vs multi-future+consistency; (ii) varying K ; (iii) varying λ_{cons} ; (iv) alternative definitions of $\mathcal{C}$. Collect qualitative case studies (occlusion, near-crosswalk ambiguity).
- **Week 7-8: Consolidation + website update + final report & presentation** Consolidate final tables/figures and error analysis. Update the project website with final results, plots, and qualitative videos/figures. Prepare slides for the final presentation.

References

- Rasouli et al., “PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction,” ICCV 2019.
- PIEPredict repository: <https://github.com/aras62/PIEPredict>